

PicoScope

Custom Oscilloscope Project

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Project Overview

The PicoScope is a compact, low-cost digital oscilloscope designed and built by our team for embedded systems testing and signal analysis. The project went through two major development iterations, each improving on hardware capability, display quality, and measurement accuracy. Both versions share the same core operating principle: a voltage divider probe circuit scales the incoming signal into a safe ADC-readable range, protection diodes guard the microcontroller inputs, and firmware samples the analog waveform in real time to render it on a local display.

Operating Principle

Both versions are built around the same fundamental signal chain:

- **Probe Input** – the signal under test is fed through a probe tip directly into the front-end circuit.
- **Voltage Divider** – a resistive divider scales the incoming voltage down to the safe input range of the microcontroller's ADC (0–3.3 V or 0–5 V depending on version).
- **Protection Diodes** – clamping diodes at the ADC pin prevent over-voltage and reverse-voltage damage, protecting the microcontroller from transient spikes.
- **ADC Sampling** – the microcontroller reads the scaled voltage at high speed, building up a time-domain waveform sample buffer.
- **Display Rendering** – sampled data is processed and drawn as a waveform on the onboard display, with calculated readings for voltage, frequency, and period.

Version 1 – Arduino Nano Edition

Version 1 was the proof-of-concept build, designed to validate the core circuit theory and firmware approach using widely available, low-cost components. Despite the hardware constraints of the Arduino Nano, it successfully demonstrated real-time waveform capture and display.

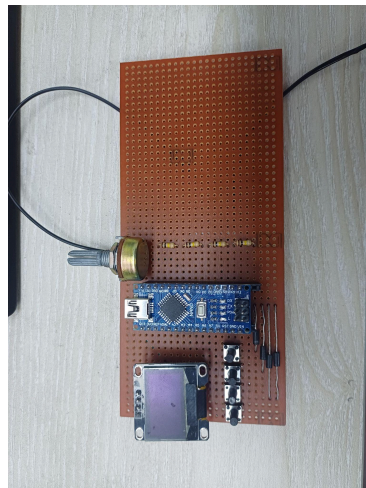
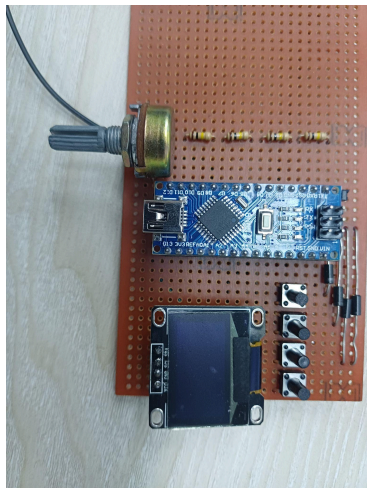
Hardware Specifications

Microcontroller	Arduino Nano (ATmega328P, 16 MHz)
ADC Resolution	10-bit (0–1023 steps)

ADC Reference Voltage	5 V (internal AVcc)
Operating Voltage	5 V USB powered
Display	0.96" OLED (128×64 px, I2C, SSD1306 driver)
Display Interface	I2C (SDA / SCL)
Input Probe Circuit	Resistive voltage divider (×10 attenuation)
Input Protection	Clamping diodes on ADC pin (D1 & D2)
Max Safe Input	~50 V peak (with ×10 divider)
Sampling Rate	~8–10 kSps (software-limited)
Bandwidth	~4 kHz estimated
Form Factor	Breadboard / perfboard prototype

Features

- Real-time waveform rendering on OLED screen
- Voltage measurement: peak-to-peak, max, and min
- Approximate frequency and period readout
- Voltage divider probe for extended input range
- Hardware over-voltage protection via diode clamping
- USB-powered via Arduino Nano onboard regulator
- Compact and portable form factor



Limitations

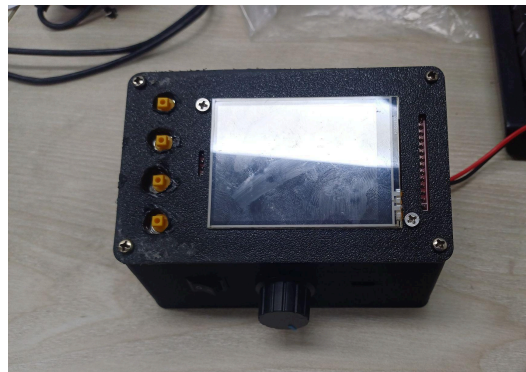
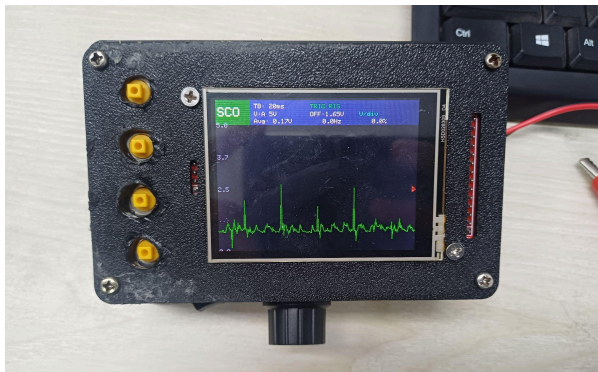
- Limited ADC speed restricts measurable frequency to low kHz range
- Small OLED display constrains waveform detail
- No hardware trigger — waveform display may roll without stable sync
- 10-bit ADC limits voltage resolution

Version 2 – Raspberry Pi Pico 2 Edition

Version 2 is a significant hardware upgrade built around the Raspberry Pi Pico 2, delivering higher ADC performance, a larger full-color TFT display, and improved signal fidelity. The front-end circuit theory remains the same — voltage divider scaling and diode protection — but the improved ADC and CPU unlock faster sampling and richer on-screen data.

Hardware Specifications

Microcontroller	Raspberry Pi Pico 2 (RP2350, dual-core Cortex-M33, 150 MHz)
ADC Resolution	12-bit (0–4095 steps)
ADC Reference Voltage	3.3 V
Operating Voltage	3.3 V logic / 5 V USB input
Display	TFT LCD color display (ST7735 / ILI9341 driver, SPI)
Display Interface	SPI (high-speed, up to 62.5 MHz)
Input Probe Circuit	Resistive voltage divider ($\times 10$ attenuation)
Input Protection	Clamping diodes on ADC pin
Max Safe Input	~ 33 V peak (with $\times 10$ divider, 3.3 V ADC ref)
Sampling Rate	~ 500 kSps (hardware ADC DMA-assisted)
Bandwidth	~ 200 kHz estimated
Form Factor	PCB / compact enclosure prototype



Features

- High-speed real-time waveform on color TFT display
- 12-bit ADC for improved voltage resolution and accuracy
- Voltage measurements: peak-to-peak, RMS (estimated), max, min
- Frequency, period, and duty cycle readout
- Color-coded waveform rendering for improved readability
- Same voltage divider and diode protection front-end as V1
- Dual-core RP2350: one core for sampling, one for display rendering

- USB-powered via Pico 2 onboard 3.3 V LDO
- Significantly higher bandwidth than Version 1

Version Comparison

Specification	Version 1 (Arduino Nano)	Version 2 (Pico 2)
MCU	ATmega328P @ 16 MHz	RP2350 @ 150 MHz
ADC Resolution	10-bit	12-bit
Sampling Rate	~8–10 kSps	~500 kSps
Bandwidth	~4 kHz	~200 kHz
Display	0.96" OLED (mono)	TFT LCD (color)
Display Interface	I2C	SPI
Max Safe Input	~50 V pk	~33 V pk
Probe Circuit	Voltage divider + diodes	Voltage divider + diodes
Power	5 V USB	5 V USB
Waveform Color	Monochrome	Full Color

Conclusion

The PicoScope project successfully delivered a functional, two-version custom oscilloscope built from the ground up by the team. Version 1 established the core circuit concept and firmware approach using the Arduino Nano and OLED display, proving viability at low cost. Version 2 scaled this foundation onto the more capable Raspberry Pi Pico 2 platform, achieving a major leap in sampling rate, ADC resolution, and display quality with the introduction of a color TFT screen.

Both versions retain the same proven front-end design — a resistive voltage divider for safe input scaling and clamping diodes for microcontroller protection — demonstrating a robust, repeatable hardware architecture that can be further developed in future revisions.